Lecture 10: LTI Resistive Circuits and Equivalent One-Ports

EE201 – Circuit Theory I

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current voltage a part of a circuit (box N)

Review: equations from the graph alone

	incidence / mesh	**fundamental cutset / loop**
KCL	$A\mathbf{i} = \mathbf{0}$	$Q\mathbf{i} = \mathbf{0}$
KVL	$M\mathbf{v} = \mathbf{0}$	$B\mathbf{v} = \mathbf{0}$
$n-1$ voltages	node voltages $\mathbf{e}$	tree voltages $\mathbf{v}_T$
$b-n+1$ currents	mesh currents $\mathbf{i}_M$	co-tree currents $\mathbf{i}_C$
works for	M: planar, nonseparable	any connected graph

KCL and KVL depend only on the graph: they hold for any elements.

A circuit has $2b$ unknowns: b more equations come from the elements.

Why this lecture?

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  - substitution theorem, equivalent one-ports
  - series and parallel connections, source transformation

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  - series and parallel connections, source transformation
- **Later — the circuit as it is:** node voltage and mesh current analysis

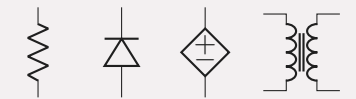
Roadmap

1. **LTI resistive circuits** — what the words mean, and what we gain
2. **A first example** — simplify a circuit step by step
3. **Substitution theorem** — replace a part of a circuit by a source
4. **One-ports and equivalence** — the same i–v characteristic at the port
5. **Series and parallel** — combine resistors and sources
6. **One source and one resistor** — exchange a voltage source and a current source

Resistive and dynamic elements

Resistive element: no memory

Its law relates v and i at the same instant, algebraically.

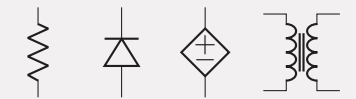


later: the ideal op-amp

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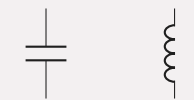
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Dynamic element: memory

Its law has a derivative.

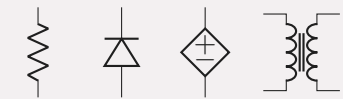


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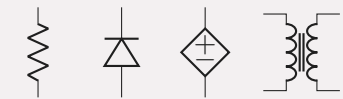
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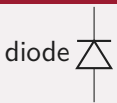
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"Resistive" means memoryless, not only "a resistor".

An element is linear if its law is linear in v, i; time-invariant if its law does not change with time.

Three questions about a circuit

Linear?

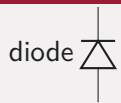


Nonlinear if it contains at least one nonlinear element, e.g. a diode.

Otherwise **linear**.

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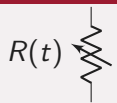
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Time-invariant?

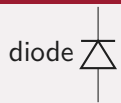


Time-varying if it contains at least one time-varying element, e.g. $R(t)$.

Otherwise **time-invariant**.

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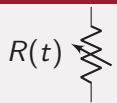
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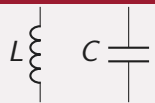
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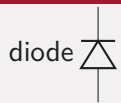


Dynamic if it contains at least one dynamic element: an L or a C.

Otherwise **resistive**.

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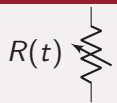
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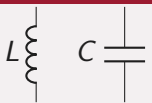
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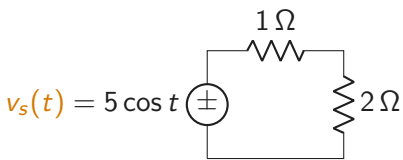
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LTI resistive circuit = linear + time-invariant + resistive:

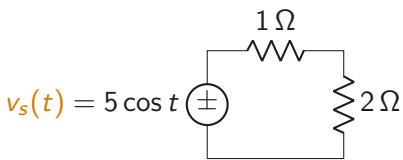
apart from the independent sources, every element is linear, time-invariant and resistive.

Independent sources are inputs



The source varies in time. Is the circuit time-varying?

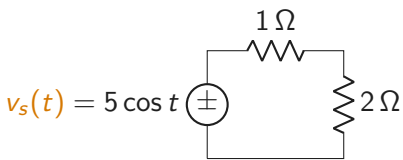
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No. An independent source is an external input: a given signal of time.

The three questions are about the circuit that the inputs drive, not about the inputs. This is an LTI resistive circuit.

What varies in time?

In every circuit $v(t)$ and $i(t)$ are functions of time. The question is whether the element law changes with time.

Time-invariant resistor

$$v(t) = R\, i(t)$$

the same rule at every instant

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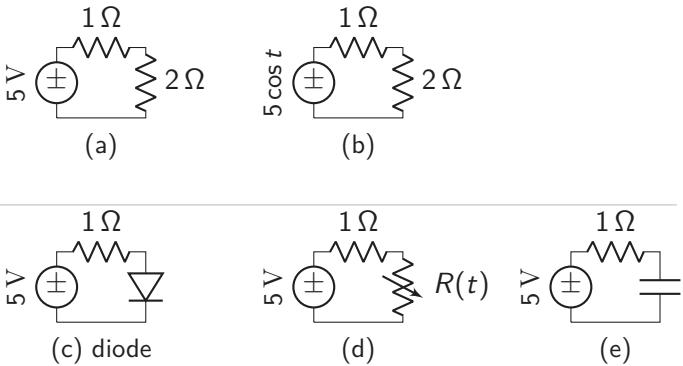
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Linear or nonlinear is also a property of the element law: $v = Ri$ is linear, a diode law is not.

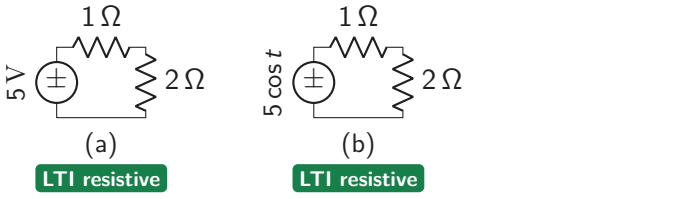
Which circuit is which?



A source, a $1\,\Omega$ resistor and one more element. In (d), $R(t) = 2 + \sin t\ \Omega$.

Linear? Time-invariant? Resistive?

Which circuit is which?



The input does not change the class: (a) and (b) are both LTI resistive.

Each of (c), (d), (e) breaks exactly one of the three properties.

What we gain

Always — no assumption on the elements

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Because the circuit is LTI resistive

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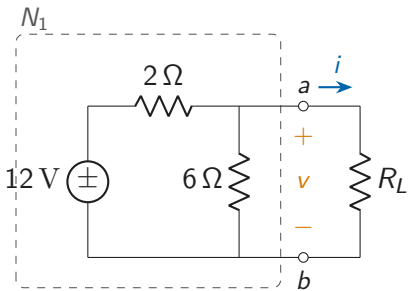
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- Time enters only through the values of the independent sources.
- So at each instant the circuit is solved like a circuit with constant sources: $2b$ linear algebraic equations.

Roadmap

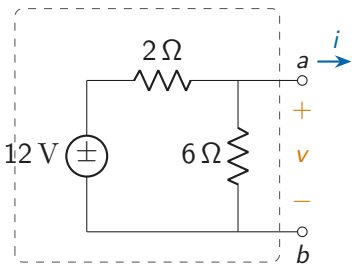
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A first example: what does R_L see?



R_L (the load) can be any resistive element.
Find v and i.

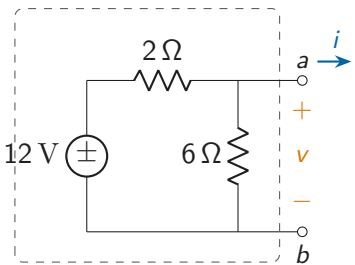
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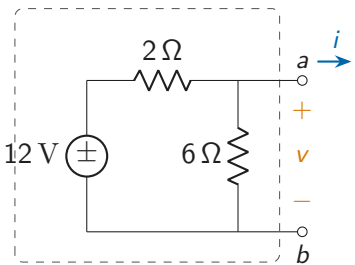


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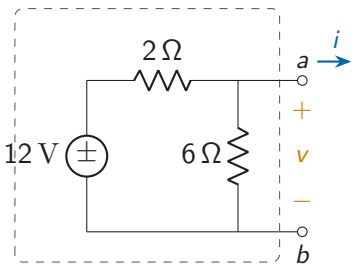
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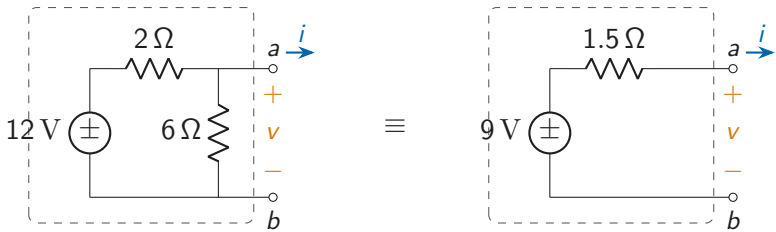
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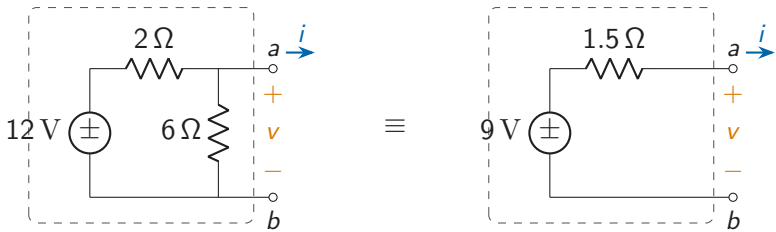
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Read the result as a circuit

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The right one obeys $v = 9 - 1.5\,i$ too (KVL): the same law.

Any load: $v = R_L i$, so $i = \dfrac{9}{1.5 + R_L}$. For $R_L = 3\,\Omega$: $i = 2\,\mathrm{A}$, $v = 6\,\mathrm{V}$.

What did we just do?

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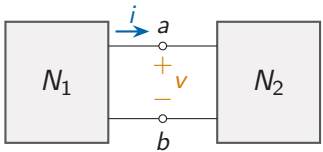
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question	answered by
May a part be replaced by a source?	substitution theorem
When may a part be replaced by a simpler part?	equivalent one-ports
Which simple replacements are there?	series, parallel, source transformation

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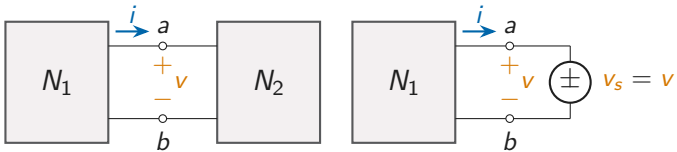
The substitution theorem



(I)

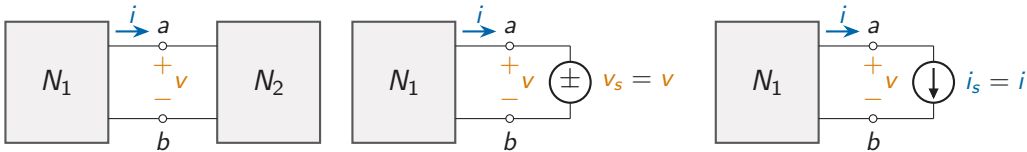
A circuit cut into two parts, joined at the terminals a, b.
i: current from N_1 into N_2 v: voltage between a and b

The substitution theorem



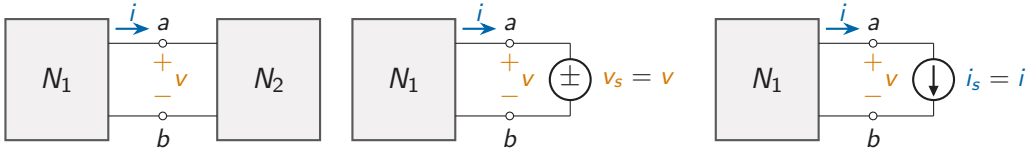
Replace N_2 by a voltage source with $v_s(t) = v(t)$ ...

The substitution theorem



. . . or by a current source with $i_s(t) = i(t)$.

The substitution theorem



Substitution theorem

No branch voltage or branch current inside N_1 changes. An observer inside N_1 cannot tell (I), (II) and (III) apart.

Condition: each circuit has a unique solution.

A unique solution

A circuit with n nodes and b branches has $2b$ unknowns, $v_1, \ldots, v_b$ and $i_1, \ldots, i_b$, and $2b$ equations:

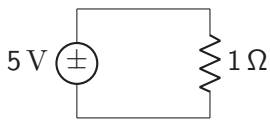
KCL	$A\mathbf{i} = 0$	$n-1$
KVL	$B\mathbf{v} = 0$	$b-n+1$
element laws	one for each branch	b

Definition

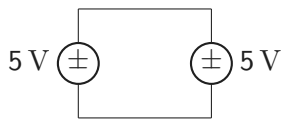
The circuit has a unique solution if exactly one $(\mathbf{v}, \mathbf{i})$ satisfies all $2b$ equations.

LTI resistive: the $2b$ equations are linear, and the solution is unique when their coefficient matrix is invertible.

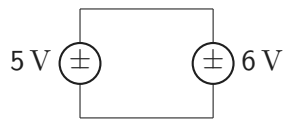
A unique solution? Three small circuits



(a)

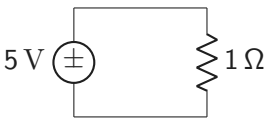


(b)



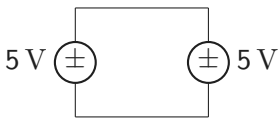
(c)

A unique solution? Three small circuits



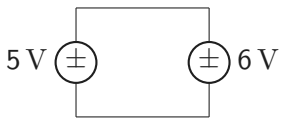
(a)

✓ unique: $i = 5\,\mathrm{A}$



(b)

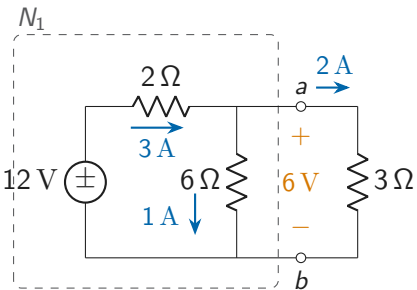
× many: any current



(c)

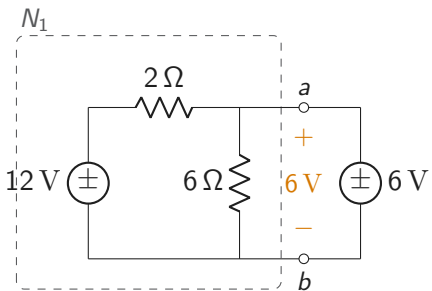
× none: KVL fails

(I) and (II): an example



(I) The first example with $R_L = 3\,\Omega$:
$v = 6\,\mathrm{V}$, $i = 2\,\mathrm{A}$; $3\,\mathrm{A}$ in the $2\,\Omega$, $1\,\mathrm{A}$ in the $6\,\Omega$.

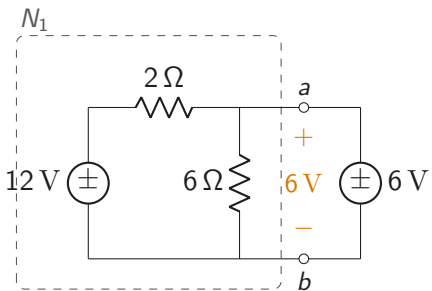
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(II) Replace R_L by a $6\,\mathrm{V}$ source. What are the values now?

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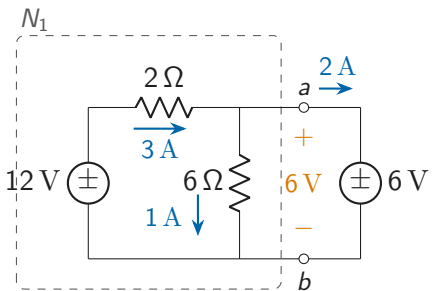


(I) The first example with $R_L = 3\,\Omega$:
$v = 6\,\mathrm{V}$, $i = 2\,\mathrm{A}$; $3\,\mathrm{A}$ in the $2\,\Omega$, $1\,\mathrm{A}$ in the $6\,\Omega$.

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KCL at a with $v = 6\,\mathrm{V}$:
$$\frac{12-6}{2} = \frac{6}{6} + i \;\Rightarrow\; i = 2\,\mathrm{A}$$

(I) and (II): an example



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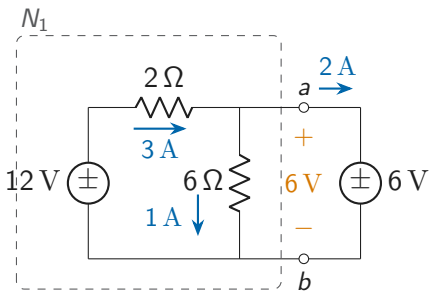
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$3\,\mathrm{A}$ in the $2\,\Omega$, $1\,\mathrm{A}$ in the $6\,\Omega$: nothing changed. ✓

(I) and (II): an example



(I) The first example with $R_L = 3\,\Omega$: $v = 6\,\mathrm{V}$, $i = 2\,\mathrm{A}$; $3\,\mathrm{A}$ in the $2\,\Omega$, $1\,\mathrm{A}$ in the $6\,\Omega$.

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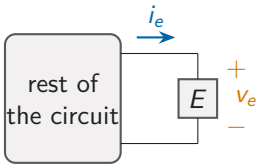
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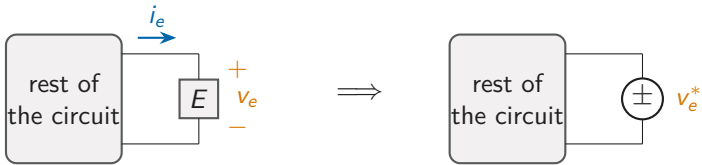
In (I) we solved the same equation: there too $v = 6\,\mathrm{V}$.

(I) and (II): one element first



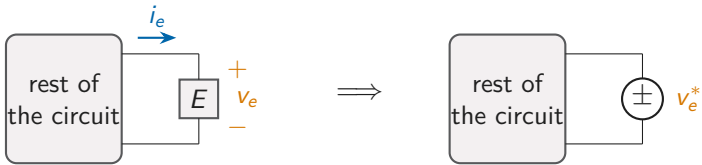
Any element E of a circuit; in the solution of (I) it has v_e^* and i_e^*.

(I) and (II): one element first



Replace only E by a voltage source. Assume both circuits have a unique solution.

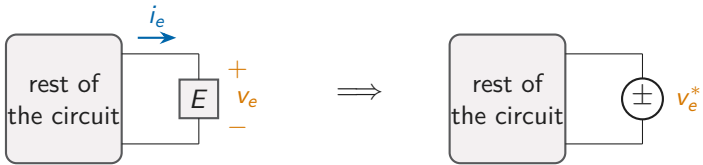
(I) and (II): one element first



Replace only E by a voltage source. Assume both circuits have a unique solution.

	(I)	(II)
KCL	$A\mathbf{i} = 0$	the same (same graph)
KVL	$B\mathbf{v} = 0$	the same (same graph)
the other $b-1$ elements	their laws	the same
element E	the law of E	$v_e = v_e^*$

(I) and (II): one element first

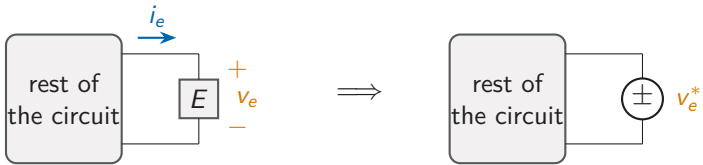


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- The solution of (I) satisfies all equations of (II). ✓

(I) and (II): one element first

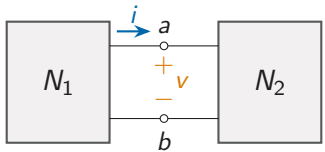


Replace only E by a voltage source. Assume both circuits have a unique solution.

	(I)	**(II)**
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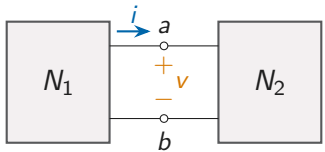
- The solution of (I) satisfies all equations of (II). ✓
- (II) has only one solution, so this is it: nothing changes, also $i_e = i_e^*$.

(I) and (II): from one element to N_2



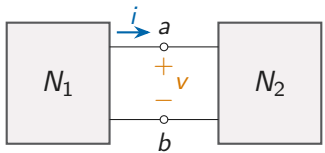
- Seen from N_1, all of N_2 acts like one branch from a to b, with v and i.

(I) and (II): from one element to N_2



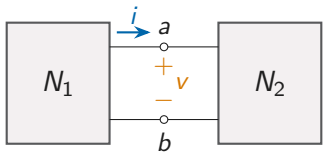
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- So take $E = N_2$: replacing it by a voltage source changes only its own equations.

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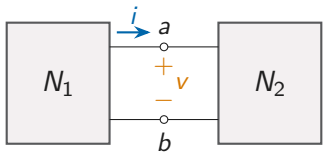
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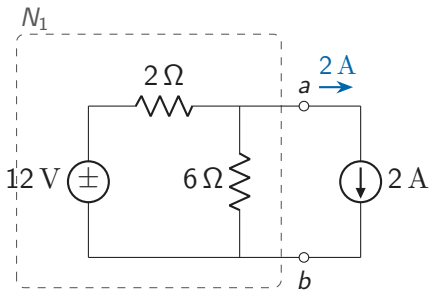
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(I) and (II): from one element to N_2



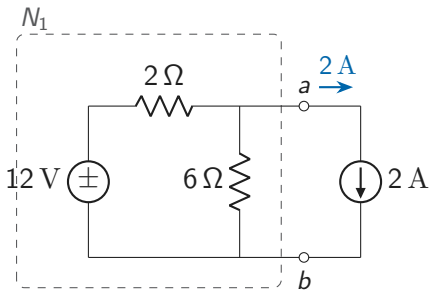
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- N_2 is removed, so it can be anything, even nonlinear.

(I) and (III): an example



(III) Replace R_L by a $2\,\mathrm{A}$ source. What are the values now?

(I) and (III): an example

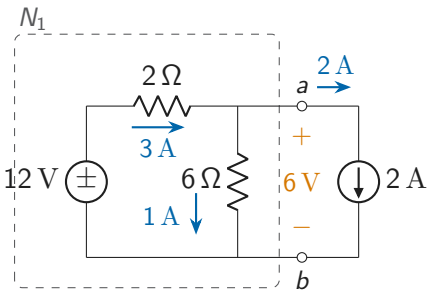


(III) Replace R_L by a $2\,\mathrm{A}$ source. What are the values now?

KCL at a with $i = 2\,\mathrm{A}$:

$$\frac{12 - v}{2} = \frac{v}{6} + 2 \;\Rightarrow\; v = 6\,\mathrm{V}$$

(I) and (III): an example



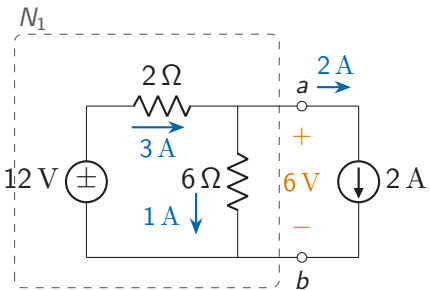
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$3\,\mathrm{A}$ in the $2\,\Omega$, $1\,\mathrm{A}$ in the $6\,\Omega$: nothing changed. ✓

(I) and (III): an example



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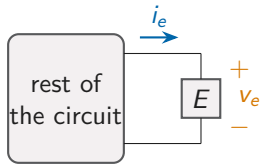
KCL at a with $i = 2\,\mathrm{A}$:

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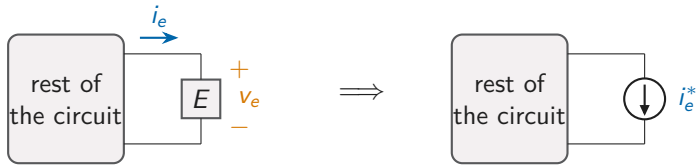
In (I) this equation holds too: there $i = 2\,\mathrm{A}$.

(I) and (III): one element first



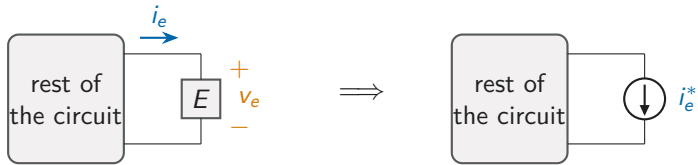
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(I) and (III): one element first



Replace only E by a current source. Assume both circuits have a unique solution.

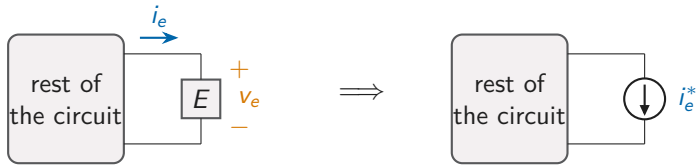
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	(I)	(III)
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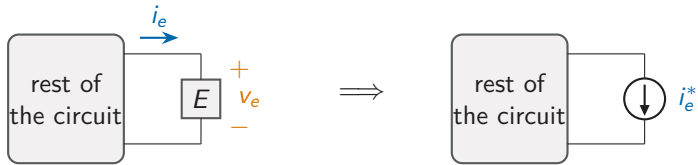


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- The solution of (I) satisfies all equations of (III). ✓

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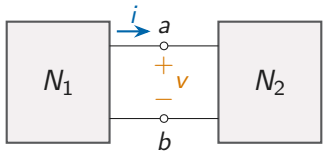


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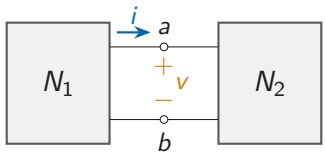
- The solution of (I) satisfies all equations of (III). ✓
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(I) and (III): from one element to N_2



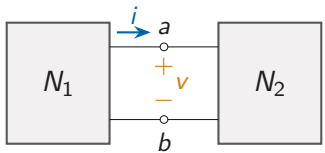
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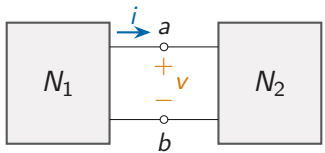
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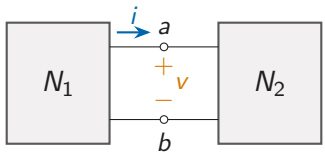
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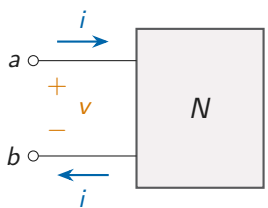
Roadmap

1. **LTI resistive circuits** — what the words mean, and what we gain ✓
2. **A first example** — simplify a circuit step by step ✓
3. **Substitution theorem** — replace a part of a circuit by a source ✓
4. **One-ports and equivalence** — the same i–v characteristic at the port
5. **Series and parallel** — combine resistors and sources
6. **One source and one resistor** — exchange a voltage source and a current source

Ports and one-ports

Port, one-port

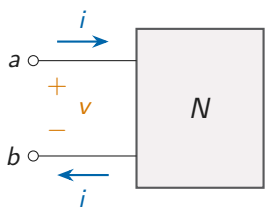
A pair of terminals is a port if the current entering one terminal equals the current leaving the other. A one-port is a circuit, or a circuit element, with one pair of terminals accessible from outside.



Ports and one-ports

Port, one-port

A pair of terminals is a port if the current entering one terminal equals the current leaving the other. A one-port is a circuit, or a circuit element, with one pair of terminals accessible from outside.

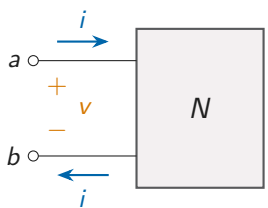


- A one-port is always a port: KCL on a closed surface around N.

Ports and one-ports

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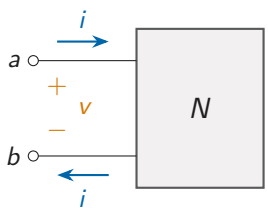


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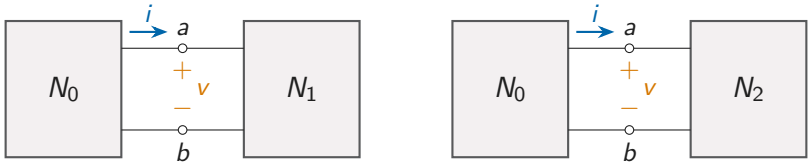


- A one-port is always a port: KCL on a closed surface around N.
- i enters the $+$ terminal of v.
- The relation between v and i is the i–v characteristic, e.g. $v = Ri$ for a resistor.

Equivalent one-ports

Equivalent one-ports

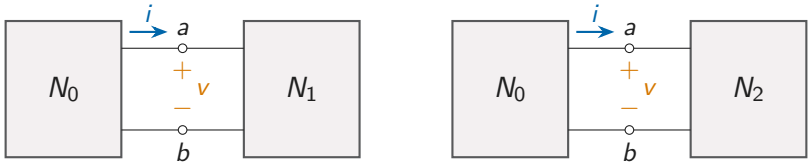
$N_1 \equiv N_2$ if they have the same i–v characteristic at their terminals.



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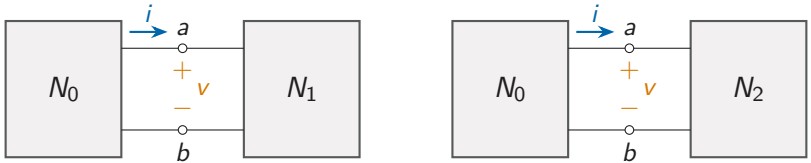


For any N_0: every branch voltage and current inside N_0 is the same in both.

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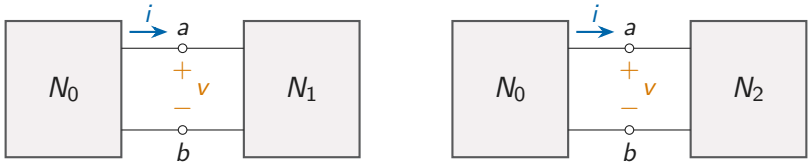
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Why: the equations are those of N_0 plus the i–v characteristic at the port.

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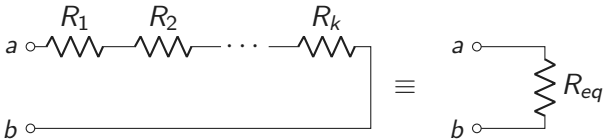
Inside N_1 and N_2 the branch values may differ.

Roadmap

1. **LTI resistive circuits** — what the words mean, and what we gain ✓
2. **A first example** — simplify a circuit step by step ✓
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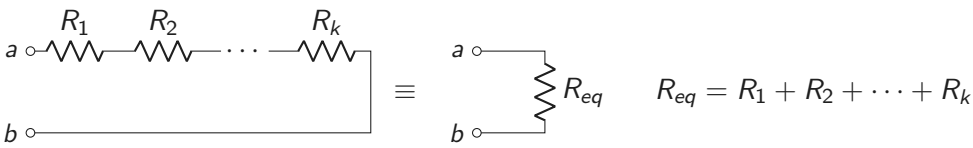
Series connection

Same current i through every element (KCL); the voltages add (KVL).



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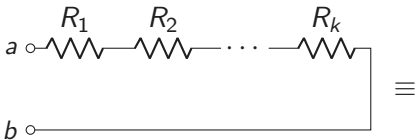


$$R_{eq} = R_1 + R_2 + \cdots + R_k$$

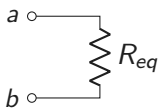
$$v = R_1 i + R_2 i + \cdots + R_k i = R_{eq} i$$

Series connection

Same current i through every element (KCL); the voltages add (KVL).



$\equiv$

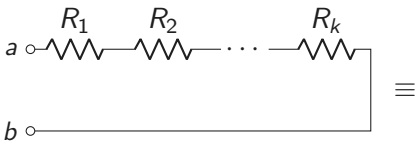


$$R_{eq} = R_1 + R_2 + \cdots + R_k$$

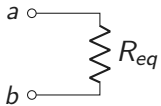
Voltage divider: the share of R_j is $v_j = \dfrac{R_j}{R_{eq}}\, v$

Series connection

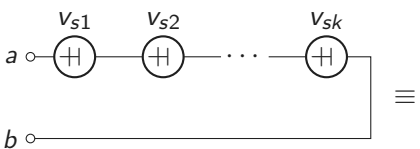
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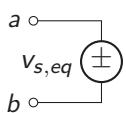
$\equiv$



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$\equiv$

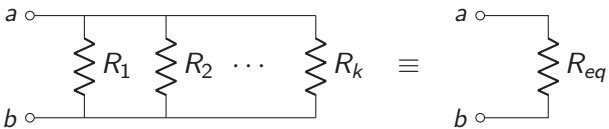


$$v_{s,eq} = v_{s1} + v_{s2} + \cdots + v_{sk}$$

All $+$ signs towards a; a source turned the other way enters with $-$.

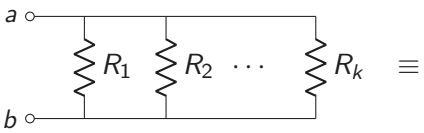
Parallel connection

Same voltage v across every element (KVL); the currents add (KCL).



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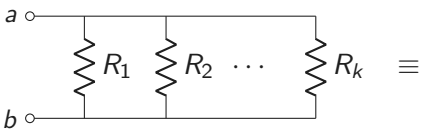


$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots + \frac{1}{R_k}$$

With $G_j = 1/R_j$: $\quad i = G_1 v + \cdots + G_k v = G_{eq} v, \quad G_{eq} = G_1 + \cdots + G_k$

Parallel connection

Same voltage v across every element (KVL); the currents add (KCL).

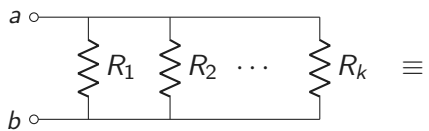


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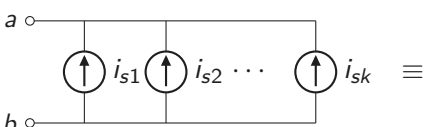
Current divider: $i_j = \dfrac{G_j}{G_{eq}}\, i.$ Two resistors: $R_{eq} = \dfrac{R_1 R_2}{R_1 + R_2}$

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Same voltage v across every element (KVL); the currents add (KCL).



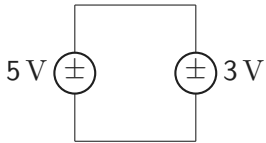
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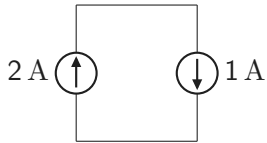
$$i_{s,eq} = i_{s1} + i_{s2} + \cdots + i_{sk}$$

All arrows towards a; a source turned the other way enters with $-$.

Two connections to avoid



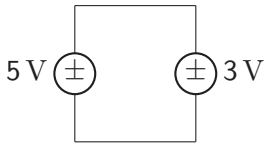
two voltage sources in parallel



two current sources in series

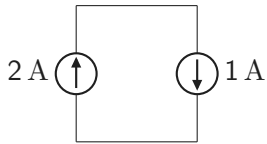
Are these allowed?

Two connections to avoid



two voltage sources in parallel

KVL fails: $5 \neq 3$.



two current sources in series

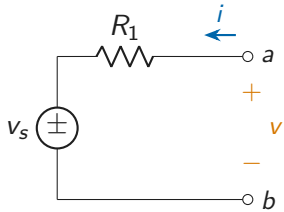
KCL fails: $2 \neq 1$.

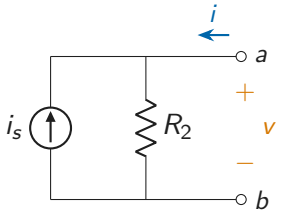
With equal values they are allowed, but the shared current (or voltage) is not fixed.

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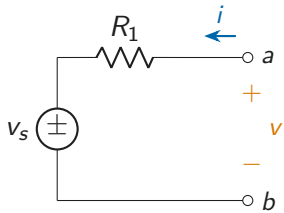
One source and one resistor

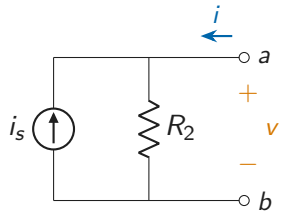




When do these two one-ports have the same i–v characteristic?

One source and one resistor

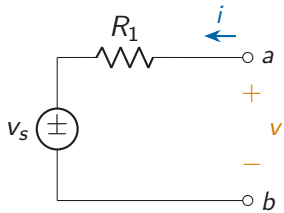




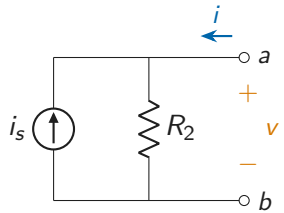
$$v = R_1 i + v_s$$

KVL around the loop: R_1 carries i.

One source and one resistor



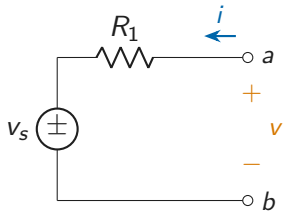
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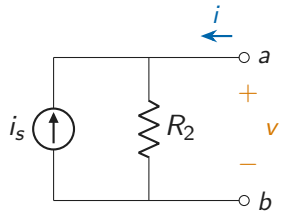
$$v = R_2(i + i_s) = R_2 i + R_2 i_s$$

i and i_s both flow into the top node: R_2 carries $i + i_s$.

One source and one resistor



$$v = R_1 i + v_s$$



$$v = R_2(i + i_s) = R_2 i + R_2 i_s$$

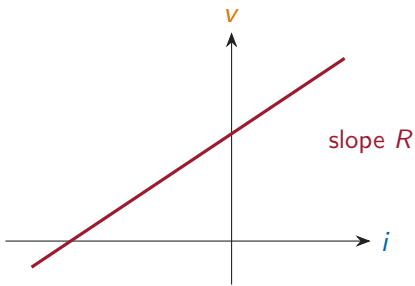
Both are straight lines in the i–v plane: equal slopes and equal intercepts.

Source transformation

Source transformation

The two one-ports are equivalent if

$$R_1 = R_2 = R \qquad \text{and} \qquad v_s = R\, i_s.$$



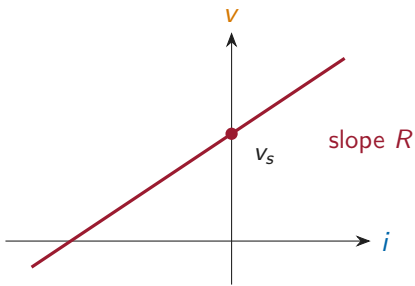
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Open circuit ($i = 0$): $v = v_s$.



Source transformation

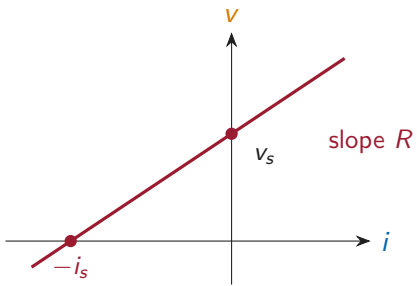
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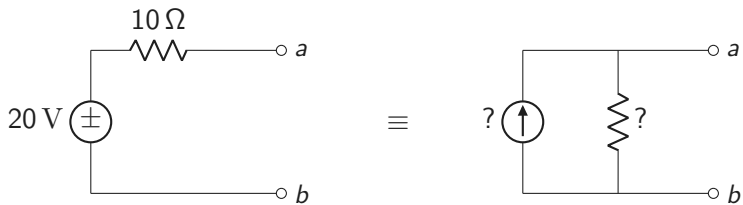
$$R_1 = R_2 = R \qquad \text{and} \qquad v_s = R\, i_s.$$

Open circuit ($i = 0$): $v = v_s$.

Short circuit ($v = 0$): $i = -i_s$.

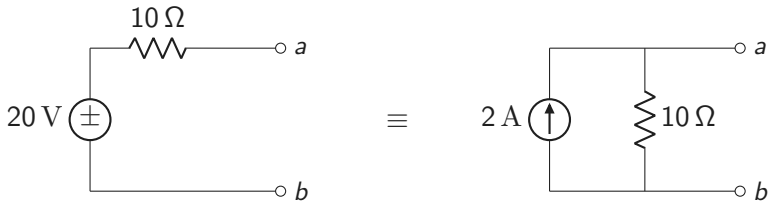


Example: a source transformation



Find the current source and the resistor.

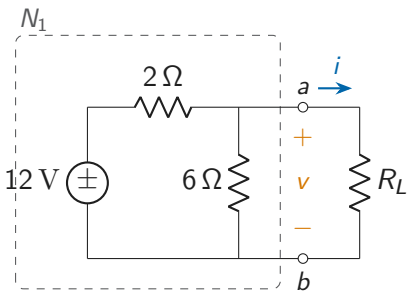
Example: a source transformation



$$R = 10\,\Omega, \qquad i_s = v_s/R = 20/10 = 2\,\mathrm{A}$$

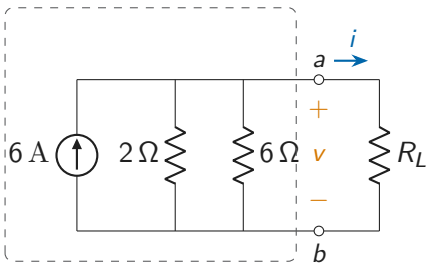
Both: open-circuit voltage $20\,\mathrm{V}$, short-circuit current $2\,\mathrm{A}$ (out of a).

The first example, again



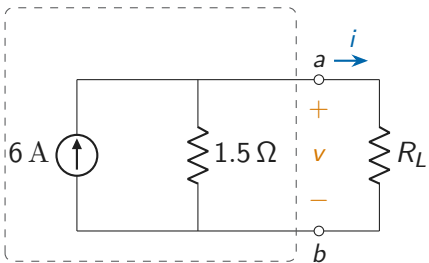
Now simplify only the part left of a–b; R_L stays.

The first example, again



Step 1. Source transformation:
$12\,\mathrm{V}$ with $2\,\Omega \rightarrow 6\,\mathrm{A}$ with $2\,\Omega$ $(12 = 2 \times 6)$

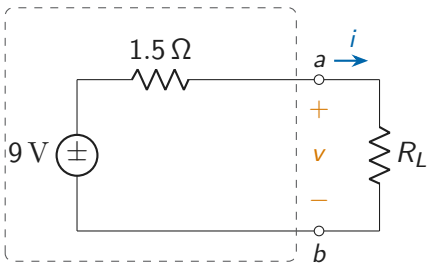
The first example, again



Step 1. Source transformation:
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Step 2. Parallel: $2\,\Omega \parallel 6\,\Omega = 1.5\,\Omega$

The first example, again

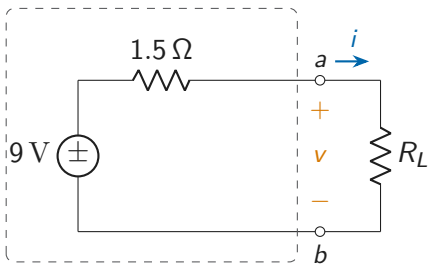


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Step 3. Source transformation:
$6\,\mathrm{A}$ with $1.5\,\Omega \rightarrow 9\,\mathrm{V}$ with $1.5\,\Omega$
$(9 = 1.5 \times 6)$

The first example, again



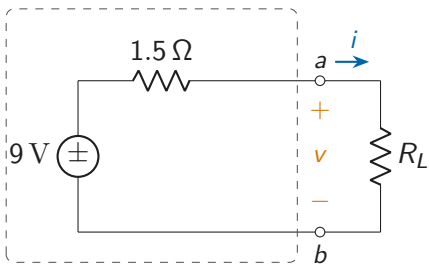
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$6\,\mathrm{A}$ with $1.5\,\Omega \rightarrow 9\,\mathrm{V}$ with $1.5\,\Omega$
$(9 = 1.5 \times 6)$

Same $9\,\mathrm{V}$ and $1.5\,\Omega$, now by transforming the circuit.

The first example, again



Step 1. Source transformation:
$12\,\mathrm{V}$ with $2\,\Omega \to 6\,\mathrm{A}$ with $2\,\Omega$ $(12 = 2 \times 6)$

Step 2. Parallel: $2\,\Omega \parallel 6\,\Omega = 1.5\,\Omega$

Step 3. Source transformation:
$6\,\mathrm{A}$ with $1.5\,\Omega \to 9\,\mathrm{V}$ with $1.5\,\Omega$
$(9 = 1.5 \times 6)$

Same $9\,\mathrm{V}$ and $1.5\,\Omega$, now by transforming the circuit.

Careful: inside, the values differ; for $R_L = 3\,\Omega$ the $12\,\mathrm{V}$ source carries $3\,\mathrm{A}$, the $9\,\mathrm{V}$ source $2\,\mathrm{A}$.

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- Both start from the matrices met earlier (A, M), now with linear element equations.

Summary

- LTI resistive: apart from the independent sources (external inputs), every element is linear, time-invariant and memoryless, so all equations are linear and algebraic.
- Substitution: replace N_2 by a voltage source $v_s = v$ or a current source $i_s = i$; nothing changes inside N_1.
- Equivalent one-ports: the same i–v characteristic; the rest cannot tell them apart.
- Series: $R_{eq} = \sum R_j$, $v_{s,eq} = \sum v_{sj}$. Parallel: $G_{eq} = \sum G_j$, $i_{s,eq} = \sum i_{sj}$.
- Source transformation: v_s with series $R \equiv i_s = v_s/R$ with parallel R.